# Thesis Checklist

Remaining work reassessed on 3 November 2026 against the current local thesis, code, and prior checklist. Earlier citation cleanup and other sections marked 02.11 remain outside this checklist. Tasks are checks, not findings that every proposed correction is established.

## 3.2.2 Bregman and beta-divergences

- **Repair the general beta-divergence setup.** Reconcile the real-valued domain and regularity assumptions with the power generator for general real beta > 0; treat the quadratic beta = 1 case separately. [chapter 164-181]
- **Check the symmetry characterization.** Compare the stated assumptions and conclusions with the cited theorem, including the Legendre-type assumptions; establish the needed hypotheses or revise the scope. [chapter 159-163; Tsuchida and Nielsen, Theorem 12]

## 3.2 Numerical approximation

- **Verify settings of saved results.** Check the Fourier quadrature settings actually used for the older joint-limit and repeated-sampling outputs. Report verified settings rather than assuming the current default. [chapter 213-226; saved outputs and notebooks]

## 3.4.1 Fixed sample experiments

- **Reconcile the joint-limit claim.** Compare the claim for alpha > 1/2 with the figure and theory, including the N^2 mesh-scale and N^3 collapse regimes; qualify the claim to match the evidence and assumptions. [chapter 460-469, 620-627]
- **Resolve the joint-limit notes.** Decide how to explain N^2 and N^3, the requested alpha > 2 experiments, alternative grids and repeated sampling. Remove or replace the continue-here marker when finished. [chapter 458-469]

Sources: chapters/5_bandwidth_selection.tex; Code/Code_kde/notebooks/03_fixed_sample_experiments.ipynb; Tsuchida and Nielsen (2026), Theorem 12.

# Numerical experiments

## 3.4.2 Repeated sampling experiments

Remaining tasks for the two repeated-sampling comparisons.

- **Explain the two comparisons.** Describe the two lambda settings, M, n, N, the five-grid-point bins, and the ISE and energy selectors, including their search domains. [chapter 472-501; repeated-sampling notebook]
- **Describe each panel accurately.** Explain the comparison in each panel separately and check whether the claim of a large discrepancy describes both. [chapter 476-500]
- **Give the parent and right-hand panel distinct labels.** The right-hand subfigure and the parent figure still use the same LaTeX label, making references ambiguous. [chapter 494, 500]
- **Resolve the repeated-sampling note.** Decide whether further fixed-sample experiments under repeated sampling are needed, and revise or remove the remaining TODO when the decision is implemented. [chapter 474]

Sources: chapters/5_bandwidth_selection.tex; Code/Code_kde/notebooks/04_repeated_sampling_experiments.ipynb; referenced figure PDFs.

# Explanation of the results

## 3.5.1 Closest adjacent pair

Remaining checks for the closest-pair subsection.

- **Distinguish vector concentration from bandwidth selection.** The limiting Perron vector puts mass on the two grid points, while an argmax selects one. Address ties and describe the finite-grid figure without saying that one grid point converges to two distinct bandwidths. [chapter 529-535; kde_tools/perron.py]
- **Check the joint-limit figure reference.** The fixed-N, large-lambda argument does not by itself establish a joint N, lambda limit. The referenced yellow curve is labeled lambda = 1 as N varies; check whether and how the argument explains that curve. [chapter 535; joint-limit figure]
- **Finish the presentation.** Review figure and color references, the remaining red text and the subsection polishing marker. [chapter 512-535]

Sources: chapters/5_bandwidth_selection.tex; Code/Code_kde/kde_tools/perron.py; the joint-limit figure.

# Explanation of the results

## 3.5.2 Statements and main-chapter proofs

The Frechet-derivative lemma proof and row-sum proposition proof are now in the main chapter. Their former appendix headings are empty; the checklist no longer treats the commented-out proof links as missing active references.

- **Polish the Frechet-derivative introduction.** Check domain and codomain notation, the line-segment condition, and the operator-valued derivative. Explain how the assumptions yield the uniform remainder bound. [chapter 537-564]
- **Check the derivative-lemma proof.** Define dilation and translation operators consistently; check derivative notation, L2 convergence and continuity, and the uniform modulus on a compact interval. [chapter 566-611]
- **Make proposition assumptions explicit.** State the row-sum and Perron-vector setups together: lambda scaling, the grid, uniqueness of the energy minimum and any interior or boundary requirements used in the proofs. [chapter 620-627, 779-781; appendix 329 onward]
- **Check the geometric-grid case.** Experiments use r = 0, while the row-sum proposition assumes 0 < r <= 1. Check both proofs at r = 0, including formulas divided by r, or qualify the connection. [chapter 620-627; main proof 629-754; appendix 329 onward]
- **Correct and cite the Laplace-method remark.** Check the sign of the second derivative at an interior maximum, the needed nondegeneracy and other hypotheses, and the displayed formula; add a source. [chapter 758-765; NIST DLMF 2.3(iii)]
- **Clarify row sums, weights and the conclusion.** Distinguish the proved row-sum approximation from the Perron-vector statement. Specify vector normalization and distinguish weighted energy from the unweighted derivative norm. [chapter 767-781]
- **Resolve empty proof headings and polishing notes.** Decide how to handle the empty derivative-lemma and row-sum appendix subsections now that their proofs are in the main chapter. Review the inline polishing markers after the relevant proofs and propositions. [appendix 325-329; chapter 535, 777]

Sources: chapters/5_bandwidth_selection.tex; chapters/9999_appendix.tex; NIST DLMF 2.3(iii), Laplace method.

# Proof checks

## 3.5.2 Row-sum and Perron proofs

- **Row-sum proof: track scaling and uniform estimates.** Check the transformed grid, the epsilon_N limit, the uniform local remainder, Gaussian tail bound and Riemann-sum limit for an interior limiting bandwidth. [main chapter 629-754]
- **Row-sum proof: establish positivity before its use.** Make explicit why the lower-bound constant and Q-hat_r(h_0) are positive and how they support the Gaussian integral and final square-root formula. [main chapter 629-754]
- **Perron proof: define missing notation.** Introduce transformed coordinates t_i and interval endpoints, m(t), q_0, sigma_N and maximizer indices before use; keep N and subsequence dependence consistent. [appendix 329 onward]
- **Perron proof: verify uniform bounds at boundaries.** Justify local expansion and row-sum upper bounds for sequences approaching either endpoint, where the row-sum proposition states an interior limit. Verify compactness and continuity steps. [appendix, proof of Proposition main_thm2; main chapter 620-627]
- **Perron proof: repair the Rayleigh lower bound.** Check the interval J for a boundary minimizer; correct inconsistent b_N and index-set notation, inside/outside pair count, and the sign of v-u in the lower bound. Verify E_N tends to zero. [appendix, proof of Proposition main_thm2]
- **Perron proof: finish the convergence argument.** Check the transition from eigenvalue and uniform row-sum bounds to the argmax index, including uniqueness, tie convention and possible endpoint minimizers. [appendix, proof of Proposition main_thm2]

## Final check

- **Compile the finished thesis.** After substantive revisions, check figures, captions, labels, bibliography and cross-references. [thesis source and compiled PDF]

Sources: chapters/5_bandwidth_selection.tex, especially lines 620-754; chapters/9999_appendix.tex, proof of Proposition main_thm2.